The Embedded Cancel Option in Limit Orders: Valuation Across Queue Levels in a High-Frequency Equity Market

Siddhant Laddha

March 2026

Abstract. We measure the economic value of the embedded option to cancel a resting limit order across ten queue levels in a US equity limit order book. Using nineteen trading days of NASDAQ ITCH message-by-order (MBO) data for Alphabet Inc. (GOOGL) and a counterfactual dual-tracker methodology, we match each live order with a passive shadow order that is prohibited from cancelling. The difference in outcomes isolates V_{cancel} — the incremental tick-value generated by the cancel decision — at each level for both sides of the book. Cancel option value per cancel event increases monotonically from 30–33 ticks at the touch to 57–68 ticks at the tenth price level, consistent with greater adverse selection exposure at depth. Expressed as an expected contribution per order placed, cancel flexibility generates 0.5–3.3 additional ticks per order — a multiple of 8–26 times the passive fill return at each level. Critically, at the second and third price levels, the expected order value under a fully passive strategy is negative: profitable liquidity provision at these depths relies entirely on the ability to cancel. The ask and bid sides exhibit a widening asymmetry at depth, with ask-side decisions generating higher option value but lower accuracy. Repositioning — cancelling and immediately re-entering — adds +585–654 basis points of alpha at the touch but destroys value at level ten (−85 to −90 bps), providing a concrete depth threshold for reposition suppression in automated market-making strategies.

Keywords: limit order book, cancel option, adverse selection, market microstructure, queue management, counterfactual analysis, high-frequency trading.

Introduction

Limit orders dominate modern electronic equity markets. On major US exchanges, order submissions, modifications, and cancellations account for over ninety-five percent of all messages by count, with executed trades forming only a small fraction of total activity. Despite this dominance, the precise economic value embedded in the right to cancel a resting limit order has not been quantified empirically across the full depth of the book using order-level counterfactual methods.

When a trader posts a limit order, they accept an asymmetric contract: they commit to supply or demand liquidity at a fixed price while the contra party retains the right to trade or not at will. The limit order provider's principal defence against adverse selection is the right to cancel — to withdraw the order before an informed counterparty reaches it. This right is costless in most modern venues and exercised at high frequency: across our sample, cancel events occur at rates of 2–37% across levels (the L2 anomaly is discussed in Section 4.3). Yet despite its ubiquity, the economic value of this right has largely been inferred indirectly, through value-function estimates in MDP frameworks, rather than measured directly through order-level counterfactuals.

The closest antecedent to this paper is Kwan and Philip [2025], who use a reinforcement learning framework to show that cancel flexibility contributes positively to limit order value through heavy-tailed realisations, consistent with a rare but large payoff structure. The replication study by Joshi et al. [2025] extends this to GOOGL and AAPL using tabular value iteration, finding cancel contribution ratios of 256% for Google and 5,200% for Apple at a one-step horizon — ratios that, as the authors note, are inflated by a small denominator at short horizons and converge toward the original paper's ∼19% contribution at absorbing-event horizons. Our paper takes a complementary approach: rather than estimating cancel value through a value function, we measure it directly via a matched-pair counterfactual, producing a V_{cancel} estimate in ticks per cancel event that is not horizon-sensitive. We then express the contribution per order as an enhancement over passive fill returns, a framing directly comparable to practitioners' assessment of alpha decay.

Our methodology constructs a dual virtual order tracking system. For each qualifying MBO event at levels 1–10, two identical virtual orders are spawned. A standard order is subject to cancel rules driven by a three-gate Queue Pressure Ratio (QPR) signal. A shadow order is prohibited from cancelling and rests until a natural fill or session end. Both orders observe the same market conditions. The difference in terminal outcomes — fill price, fill time, or

no fill — isolates the cancel option value, order by order.

The principal empirical findings are as follows. First, V_{cancel} per cancel event rises monotonically from 30–33 ticks at level one to 57–68 ticks at level ten, consistent with greater adverse selection exposure at depth. Second, expressed as expected contribution per order placed, cancel flexibility generates 0.5–3.3 additional ticks per order — a factor of 8–26 times the passive fill return across levels. Third, at levels two and three, the expected order value under a fully passive (no-cancel) strategy is negative: cancel flexibility is existentially necessary at these depths. Fourth, a widening ask/bid asymmetry emerges at depth. Fifth, reposition alpha reverses sign between the touch (+600 bps) and level ten (−90 bps).

The paper is organised as follows. Section 2 describes the data. Section 3 presents the dual-tracker methodology. Section 4 reports empirical results. Section 5 interprets the findings. Section 6 discusses strategy implications. Section 7 acknowledges limitations. Section 8 concludes.

Data

The empirical analysis uses NASDAQ ITCH message-by-order (MBO) data for Alphabet Inc. (GOOGL, instrument identifier 7152). The sample covers nineteen trading days from February 6 to March 5, 2026 — a period of typical market activity for this security, without earnings announcements or major macroeconomic events. The instrument trades with a minimum tick size of $0.01 and a sample-period mid-price of approximately $175.

NASDAQ ITCH MBO data records every order event — submission, cancellation, modification, and execution — with nanosecond-resolution timestamps and full queue-position information for each individual order identifier. This granularity distinguishes MBO from fixed-interval LOB snapshots: the pipeline tracks the exact queue position of each virtual order as it changes event by event, a precision essential for the validity of the counterfactual. A ten-minute warm-up period at the start of each session excludes early-day events where the LOB may be incompletely reconstructed.

Across the nineteen-day sample, the pipeline processes approximately 7.78 million fills and 706,320 cancel events — a total order population of approximately 8.49 million events across ten levels on both sides. The mean fill return is 0.113 ticks with a coefficient of variation of 11% across daily means (range: 0.091–0.131 ticks), indicating a stable signal over the sample. All results are reported separately for the ask (offer) and bid sides.

Methodology

Dual Virtual Order Tracking

For each qualifying MBO event at levels 1–10, the pipeline spawns a matched pair of virtual orders that subsequently observe identical market conditions:

- **Standard order:** Subject to cancel rules. Receives QPR cancel signals, BBO-distance reposition triggers, and a 5-minute maximum lifetime.
- **Shadow order:** Passive counterfactual. Cancel rules disabled entirely. Rests until a natural fill or session end.

Both orders process the same MBO add, cancel, trade, and fill messages in nanosecond order. At most 100 standard-shadow pairs are active simultaneously. The terminal outcome of each order is recorded: fill price, fill timestamp, or an unfilled cancel. Matching outcomes by order identifier isolates the cancel option value.

Cancel Option Value

For each cancelled standard order, cancel option value is:

$$V_{\text{cancel}} = \text{RL_PnL} - \text{Shadow_PnL} \tag{1}$$

Both terms are normalised by tick size alone, making V_{cancel} size-independent. The terms are computed conditional on whether the shadow order filled.

If the shadow order filled:

$$\text{RL_PnL} = \frac{\text{cancel_mid} - \text{shadow_mid}_{T+100\text{ms}}}{\text{tick}} \tag{2a}$$

$$\text{Shadow_PnL} = \frac{\text{shadow_fill_price} - \text{birth_mid}}{\text{tick}} \tag{2b}$$

If the shadow order did not fill:

$$\text{RL_PnL} = \frac{\text{ref_mid} - \text{birth_mid}}{\text{tick}} \tag{3a}$$

$$\text{Shadow_PnL} = 0 \tag{3b}$$

A positive V_{cancel} indicates the cancel avoided an adverse outcome — the mid-price deterio-

rated further after the would-be shadow fill. A negative V_{cancel} indicates a premature cancel — the price recovered and passive holding would have been profitable.

Cancel Option Contribution Framing

We report cancel option value in three complementary framings. First, V_{cancel} per cancel event (ticks) measures the average payoff when the option is exercised. Second, the expected cancel contribution per order placed:

$$\text{Cancel contribution/order} \;=\; P(\text{cancel}) \times \mathbb{E}[V_{\text{cancel}}] \tag{4}$$

integrates over cancel frequency to express the per-order-placed value of cancel flexibility. This is directly comparable to the per-order fill return and allows the enhancement ratio:

$$\text{Enhancement ratio} \;=\; \frac{P(\text{cancel}) \times \mathbb{E}[V_{\text{cancel}}]}{\mathbb{E}[\text{fill_return}]} \tag{5}$$

to be computed at each level and side. Third, the counterfactual no-option expected value:

$$\mathbb{E}[V_{\text{no-option}}] \;=\; P(\text{fill}) \times \mathbb{E}[\text{fill_return}] \;+\; P(\text{cancel}) \times \mathbb{E}[\text{Shadow_PnL}] \tag{6}$$

identifies levels where passive holding generates negative expected returns — establishing whether the cancel option is value-enhancing or existentially necessary.

This framing differs from the MDP-based cancel contribution ratio in Kwan and Philip [2025], which uses the full discounted value function V_{uncon} as denominator. Their ∼19% estimate uses a denominator that includes all future fill opportunities from a given state, making it substantially larger than our per-order expected fill return. Our enhancement ratio in equation (5) therefore measures a more directly interpretable quantity: for every tick earned passively, how many additional ticks does cancel flexibility generate.

QPR Cancel Signal: Three-Gate Design

Cancel decisions in the standard tracker are governed by a three-gate Queue Pressure Ratio (QPR) signal. A cancel fires if and only if all three conditions hold simultaneously:

Three-Gate QPR Cancel Architecture

Gate 1 (Signal)	*Gate 2 (Depth)*	*Gate 3 (Persistence)*
$\text{QPR} \geq 2.0$	$q_{\text{frac}} < 0.20$	$\text{QPR} \geq 2.0$ for $\geq 10\,\text{ms}$

$$\text{where} \quad \text{QPR} = \frac{\sum_{50\text{ms}} \text{self-side depletion}}{\sum_{50\text{ms}} \text{contra-BBO growth} + \varepsilon}$$

Figure 1: Three-gate QPR cancel architecture. A cancel fires only when all three conditions hold simultaneously. Gate 1 measures queue pressure via a 50 ms rolling ratio. Gate 2 ensures the order is near the front of the queue. Gate 3 suppresses spoofing and momentary MBO noise.

The QPR numerator accumulates same-side depletion events (cancels, trades, fills) at the order's limit price for volume ahead of this order. The denominator accumulates ADD messages on the contra side at the current BBO. A high QPR signals potential directional adverse flow. The depth gate ($q_{\text{frac}} < 0.20$) prevents cancelling from the back of the queue. The 10 ms persistence gate eliminates spoofing and MBO flicker.

Fill Contribution and Case Definitions

For filled orders, the 100 ms mark-out contribution is:

$$\text{toxic_ticks} = \frac{\text{fill_price} - \text{mid}_{T+100\text{ms}}}{\text{order_size} \times \text{tick}} \quad \text{[buy; sign-adjusted for sell]} \tag{7}$$

The reposition contribution compares repositioned fill outcomes against passive shadow outcomes, yielding one of four case labels (A–D) defined in Appendix B. The total fill return is toxic_ticks plus the reposition contribution, used as $\mathbb{E}[\text{fill_return}]$ throughout.

Empirical Results

Cancel Option Value Profile

Table 1 reports the cancel option value profile by level and side across the full nineteen-day sample. Three regimes emerge.

Table 1: Cancel Option Value by Level and Side

Level	Side	N cancels	P(cancel)	Shad. fill%	V_{cancel} (ticks)	V_{cancel} (bps)	Just.%	Opp. cost
L1	Ask	8,070	1.9%	85.1%	**32.8**	18.75	93.6%	5.5
L1	Bid	6,701	1.6%	85.9%	**30.4**	17.35	93.3%	5.8
L2	Ask	157,696	**36.5%**	94.8%	**6.5**†	3.69	91.0%	5.6
L2	Bid	150,929	**35.9%**	93.5%	**6.0**†	3.44	90.1%	6.0
L3	Ask	48,011	11.2%	91.4%	**16.4**†	9.40	93.1%	5.3
L3	Bid	46,105	11.0%	90.1%	**14.7**†	8.42	92.5%	5.6
L4	Ask	22,656	5.3%	84.2%	**33.3**	19.04	92.5%	6.4
L4	Bid	23,350	5.6%	84.9%	**27.4**	15.64	92.4%	6.5
L5	Ask	19,080	4.4%	79.5%	**44.9**	25.66	92.0%	7.4
L5	Bid	19,750	4.7%	81.5%	**35.8**	20.43	92.0%	7.1
L6	Ask	18,617	4.3%	75.8%	**52.4**	29.92	90.4%	9.2
L6	Bid	19,653	4.7%	79.2%	**40.9**	23.39	91.8%	7.5
L7	Ask	19,636	4.6%	73.6%	**58.2**	33.27	89.4%	10.2
L7	Bid	20,764	5.0%	79.6%	**45.6**	26.06	92.2%	7.1
L8	Ask	19,412	4.5%	71.5%	**62.4**	35.66	88.6%	10.9
L8	Bid	20,902	5.0%	78.7%	**49.0**	27.98	91.7%	7.6
L9	Ask	19,779	4.6%	67.8%	**65.0**	37.14	86.8%	12.7
L9	Bid	21,751	5.2%	77.1%	**53.0**	30.30	91.4%	8.0
L10	Ask	20,976	4.9%	65.9%	**68.1**	38.90	86.0%	13.5
L10	Bid	22,482	5.3%	77.2%	**56.7**	32.38	91.6%	7.7

V_{cancel} in ticks (\$0.01 = 1 tick). Shadow fill% = fraction of cancel events where the passive shadow order filled during the session. Justified% = fraction with $V_{cancel} > 0$ (correct decision). Opp. cost% = fraction with $V_{cancel} < 0$ (premature cancel). † Levels 2–3 affected by QPR over-triggering (Section 4.3). 19-day sample, GOOGL, Feb 6 – Mar 5, 2026.

Regime 1 — Level 1 (deep out-of-the-money put). The cancel fires in only 1.6–1.9% of orders. When it fires, V_{cancel} averages 30–33 ticks — the realisation of a rare, large adverse move that the cancel successfully avoided. Decision accuracy is high: 93.3–93.6% of events are justified, with opportunity cost of 5.5–5.8%. The shadow fill rate of 85–86% confirms that in most cancel events the passive order would have eventually filled at an adverse price.

Regime 2 — Level 2 (QPR over-trigger anomaly). The cancel rate is 35.9–36.5%, approximately seven to twenty times the rate at L4 and above. This anomaly is discussed in Section 4.3.

Regime 3 — Levels 3–10 (monotone depth gradient). V_{cancel} increases smoothly from 14.7–16.4 ticks at L3 to 56.7–68.1 ticks at L10. Cancel rates stabilise at 4–11%. Shadow fill rates decline monotonically with depth (91% at L3 to 66–77% at L10), reflecting the lower probability of a passive order reaching the front of a deep queue before session end. The monotone pattern is consistent with greater adverse selection exposure at depth, where orders rest longer and are more likely to be reached only by directional informed flow.

Cancel Option Contribution per Order and Enhancement Ratio

Table 2 reports the expected cancel contribution per order placed (equation 4), the enhancement ratio (equation 5), and the no-option expected value (equation 6).

Table 2: Cancel Option Contribution per Order Placed

Level	Side	P(cancel)	$\mathbb{E}$[fill_return] (ticks)	Contrib/order (ticks)	Contrib (bps)	$\mathbb{E}[V_{\text{no-option}}]$ (ticks)	Enhancement ratio
L1	Ask	1.9%	0.0858	**0.617**	0.353	0.054	7.2×
L1	Bid	1.6%	0.0679	**0.487**	0.278	0.091	7.2×
L2	Ask	36.5%	0.1186	**2.360**	1.348	**−0.330**△	19.9×
L2	Bid	35.9%	0.1047	**2.157**	1.233	**−0.252**△	20.6×
L3	Ask	11.2%	0.1199	**1.843**	1.053	**−0.072**△	15.4×
L3	Bid	11.0%	0.1035	**1.622**	0.927	**−0.010**△	15.7×
L4	Ask	5.3%	0.1226	**1.778**	1.016	0.012	14.5×
L4	Bid	5.6%	0.1087	**1.535**	0.877	0.066	14.1×
L5	Ask	4.4%	0.1192	**1.978**	1.130	0.027	16.6×
L5	Bid	4.7%	0.1122	**1.678**	0.959	0.088	15.0×
L6	Ask	4.3%	0.1200	**2.267**	1.295	0.030	18.9×
L6	Bid	4.7%	0.1120	**1.919**	1.096	0.088	17.1×
L7	Ask	4.6%	0.1283	**2.662**	1.521	0.041	20.7×
L7	Bid	5.0%	0.1147	**2.265**	1.294	0.091	19.7×
L8	Ask	4.5%	0.1285	**2.828**	1.616	0.049	22.0×
L8	Bid	5.0%	0.1116	**2.448**	1.399	0.079	21.9×
L9	Ask	4.6%	0.1239	**2.992**	1.710	0.054	24.1×
L9	Bid	5.2%	0.1141	**2.749**	1.571	0.080	24.1×
L10	Ask	4.9%	0.1262	**3.316**	1.895	0.061	26.3×
L10	Bid	5.3%	0.1094	**3.028**	1.730	0.077	27.7×

Contrib/order $= P(\textit{cancel}) \times \mathbb{E}[V_{\textit{cancel}}]$, *in ticks and bps of mid-price (≈$175).* $\mathbb{E}[V_{\textit{no-option}}] = P(\textit{fill}) \times \mathbb{E}[\textit{fill_return}] + P(\textit{cancel}) \times \mathbb{E}[\textit{Shadow_PnL}]$. *Enhancement ratio* $= (P(\textit{cancel}) \times \mathbb{E}[V_{\textit{cancel}}]) \,/\, \mathbb{E}[\textit{fill_return}]$. △ = *negative* $\mathbb{E}[V_{\textit{no-option}}]$; *strategy unprofitable without cancel flexibility at these levels.*

The expected cancel contribution per order ranges from 0.49–0.62 ticks at level 1 to 2.75–3.32 ticks at levels 9–10. In basis points of mid-price, this corresponds to 0.28–0.35 bps at L1 rising to 1.57–1.90 bps at L9–L10.

The enhancement ratio — additional ticks per order from cancel flexibility relative to passive fill returns — ranges from 7.9× to 8.0× at level 1 and rises to 24.0×–26.6× at levels 9–10.

For every tick of passive spread capture, the cancel option generates 8–27 additional ticks in expected per-order terms.

The no-option expected value is negative at levels 2 and 3: $-0.330\,\mathrm{t}$ (L2 ask), $-0.252\,\mathrm{t}$ (L2 bid), $-0.072\,\mathrm{t}$ (L3 ask), $-0.010\,\mathrm{t}$ (L3 bid). A purely passive strategy at these depths loses money in expectation. The cancel option is not value-enhancing at L2–L3 — it is existentially necessary. This is consistent with the heavy-tailed avoidance finding in Kwan and Philip [2025] and extends it to a level-specific identification of where the option is essential versus supplementary.

The Level 2 Anomaly

Level 2 requires separate treatment. Under the current `QPR_THRESHOLD` of 2.0, the cancel rate reaches 35.9–36.5%, compared to 4–5% at L4 and above. The cause is QPR over-triggering: at the second price level, routine bid-ask imbalances push QPR above 2.0 for 10 ms without signalling genuine toxic flow. The shadow fill rate at L2 (93.5–94.8%) is the highest across all levels, confirming that the passive shadow order would have filled in most of these cancel events — but V_{cancel} per event is only 6.0–6.5 ticks because the mid-price recovery after these false-positive cancels is modest.

Despite this measurement limitation, the no-option expected value at L2 is deeply negative (-0.25 to -0.33 ticks), confirming that even an imprecise cancel signal at L2 is economically necessary. The true V_{cancel} at L2 under a correctly calibrated threshold (recommended: QPR ≥ 5.0) would be substantially higher, likely converging toward the L3–L4 range. L2 results should be read with this caveat.

Ask/Bid Asymmetry at Depth

A consistent asymmetry between the ask and bid sides widens with queue depth. At level 1, V_{cancel} differs by only 2.4 ticks (32.8 t ask vs 30.4 t bid). By level 7, the gap has widened to 12.6 ticks (58.2 t vs 45.6 t), stabilising at 11–13 ticks at L8–L10. The shadow fill rate declines faster for the ask side at depth (65.9% at L10 ask vs 77.2% at L10 bid), and the opportunity cost rises faster for asks (13.5% at L10 ask vs 7.7% for bids).

This pattern is consistent with the ask side at depth facing more episodic, aggressive adverse flow from informed buyers. Large directional buying sweeps the offer side with greater velocity than informed selling sweeps the bid, producing higher V_{cancel} per event on the ask side. However, the same flow pattern makes the QPR signal noisier on the ask side at depth, generating more false-positive cancels. This asymmetry argues for side-specific QPR thresholds at depth.

Fill Returns and Reposition Alpha

Table 3 reports fill outcomes and reposition alpha by level and side.

Table 3: Fill Returns and Reposition Alpha by Level and Side

Level	Side	N fills	Mean fill (ticks)	Passive return	Repo return	Repo alpha (bps)
L1	Ask	421,014	0.0858	0.0776	0.1430	**+654**
L1	Bid	411,616	0.0679	0.0612	0.1198	**+585**
L2	Ask	273,772	0.1186	0.1162	0.1173	**+11**†
L2	Bid	269,797	0.1047	0.1038	0.1008	**−29**†
L3	Ask	380,260	0.1199	0.1177	0.1272	**+95**
L3	Bid	372,943	0.1035	0.1011	0.1135	**+124**
L4	Ask	401,881	0.1226	0.1192	0.1312	**+120**
L4	Bid	393,130	0.1087	0.1098	0.1027	**−71**
L5	Ask	414,227	0.1192	0.1188	0.1246	**+58**
L5	Bid	401,191	0.1122	0.1106	0.1087	**−19**
L6	Ask	411,438	0.1200	0.1192	0.1254	**+62**
L6	Bid	399,679	0.1120	0.1075	0.1154	**+79**
L7	Ask	409,749	0.1283	0.1255	0.1325	**+70**
L7	Bid	397,374	0.1147	0.1106	0.1196	**+90**
L8	Ask	408,993	0.1285	0.1232	0.1369	**+136**
L8	Bid	397,295	0.1116	0.1096	0.1134	**+38**
L9	Ask	409,813	0.1239	0.1244	0.1277	**+33**
L9	Bid	397,915	0.1141	0.1103	0.1141	**+38**
L10	Ask	409,603	0.1262	0.1321	0.1236	**−85**
L10	Bid	398,217	0.1094	0.1140	0.1050	**−90**

Passive return = mean fill return for case D fills (no reposition). Repo return = mean fill return for case A fills (re-entered after cancel). Repo alpha = repo return minus passive return, in basis points. Green = positive alpha; red = negative alpha. † *L2 figures affected by QPR over-triggering.*

Reposition alpha is strongly positive at level 1 (+654 bps ask, +585 bps bid). At L1, only 12–13% of orders are repositioned — the decision is highly selective — and re-entry captures a substantially better queue position. Across levels 3–9, ask-side repo alpha is consistently positive (33–136 bps), peaking at L8. Bid-side alpha is positive at most levels but negative

at L4 (-71 bps) and L5 (-19 bps), reflecting QPR cancels firing during rallies where bid re-entry is at a lower (worse) price.

Level 10 is the critical reversal point. Repo alpha is -85 bps (ask) and -90 bps (bid), and the reposition rate exceeds 50% on both sides — more than half of all L10 fills occur after a cancel-and-reposition. When the reposition rate exceeds 50%, the signal has lost its selectivity: repositioning has become structural churn. Passive fills at L10 (0.132 t ask, 0.114 t bid) outperform repositioned fills (0.124 t, 0.105 t), providing a clear quantitative case for reposition suppression at depth.

Economic Interpretation

The results support a unified economic interpretation. A resting limit order is a portfolio of two components: a passive fill claim worth $\mathbb{E}[\text{fill_return}]$ per fill, and an embedded cancel option worth V_{cancel} per exercise. Our data show that the cancel option, expressed as a per-order contribution (equation 4), exceeds the passive fill return by a factor of 8–27 across levels. However, this does not mean the cancel option is more important than fills in a strategic sense — fills are the primary revenue event, and cancel events exist to protect fill revenue from adverse selection erosion. The correct interpretation is that without cancel flexibility, a significant share of fill revenue would be destroyed by adverse fills that the cancel option prevents.

The existential finding at levels 2 and 3 provides the clearest illustration. At these levels, resting passively generates negative expected returns — the adverse selection hazard is large enough relative to spread capture that fills on average lose money when no cancel is possible. The cancel option transforms a structurally loss-making position into a profitable one. This is qualitatively consistent with the Kwan and Philip [2025] finding that cancellation adds value through heavy-tailed avoidance, and extends that finding to a level-specific identification of where the option is essential versus supplementary.

The reposition result is also interpretable within this framework. At the touch, the cancel option creates value through both loss avoidance (V_{cancel} per event) and position improvement (positive reposition alpha) — two channels that are complementary and both active. At extreme depth, the position improvement channel reverses sign as the reposition rate rises beyond the selectivity threshold and the signal's false-positive rate increases. The cancel option continues to generate loss-avoidance value at L10 (V_{cancel} of 57–68 ticks per event), but the reposition execution of the option destroys value.

Strategy Design Implications

Level-specific QPR calibration. The current uniform `QPR_THRESHOLD` of 2.0 over-triggers at level 2 (cancel rate 36% vs 4–5% elsewhere) and likely at level 3. Recommended thresholds: QPR ≥ 5.0 at L2, ≥ 3.0 at L3, ≥ 2.0 at L4 and above. Level-specific thresholds are standard practice in production limit order management and are well-motivated by the heterogeneous order flow dynamics at different depths identified in this analysis.

Reposition suppression at depth. Reposition alpha is negative at L10 (-85 to -90 bps) and the reposition rate exceeds 50%. The most direct fix is to increase the BBO-distance reposition threshold from 10 to 15 ticks for levels ≥ 7, requiring a larger price dislocation before a reposition fires. An alternative is to disable repositioning entirely at levels 9–10. The data at L10 support the stronger intervention: case B fills ($\sim$4% of events) indicate priority loss from repositioning, and passive returns consistently exceed repositioned returns.

Side-specific QPR thresholds at depth. The widening ask/bid asymmetry in opportunity cost at levels 7–10 (ask: 10–14%, bid: 7–8%) argues for a higher ask-side QPR threshold at depth. A threshold of 3.0 for ask-side orders at L7–L10 (vs 2.0 for bids) would reduce false-positive cancels on the offer side while preserving the more accurate bid-side signal.

Limitations and Future Research

Single instrument and short sample. The analysis covers one large-capitalisation equity (GOOGL) over nineteen trading days — a high-liquidity, penny-tick instrument in a relatively benign volatility regime. Cancel option values and enhancement ratios measured here are specific to this context. Small-cap equities with wider spreads, or instruments in high-volatility regimes (earnings periods, macro events), may exhibit substantially different dynamics. Extension to a cross-section of equities and a minimum six-month sample spanning at least one earnings cycle is the most important methodological priority for future work.

Single cancellation policy. The QPR three-gate signal is one specification of a cancel rule. The dual-tracker counterfactual framework is policy-agnostic and can accommodate any cancel specification, including machine-learning classifiers trained to predict V_{cancel} di-

rectly. Future work could compare the QPR baseline presented here against a learned policy, measuring whether additional model complexity generates higher option value and lower opportunity cost.

Relationship to MDP-based estimates. Our enhancement ratio (8–27 times over fill returns) is substantially larger than the ∼19% cancel contribution reported in Kwan and Philip [2025] and the 256% reported for GOOGL in Joshi et al. [2025]. These differences are methodological: MDP-based estimates use a value function denominator that includes all future fill opportunities from a given state, making V_{uncon} large relative to our per-order fill return. Both framings are internally consistent; they measure different things. Appendix C provides a formal reconciliation.

Zero market impact and latency. Virtual order tracking assumes simulated orders do not affect the book — reasonable at individual-order scale in GOOGL. The QPR signal executes at zero latency; live systems face 1–100 μs delays. Synthetic latency experiments would quantify the microsecond decay of cancel option value, a parameter directly relevant to co-location investment decisions.

Conclusion

This paper measures the economic value of the embedded cancel option in limit orders using a counterfactual dual-tracker methodology applied to message-by-order data. Matching each live order with a passive shadow order that cannot cancel, we isolate V_{cancel} at each of ten queue levels and on both sides of the order book.

Cancel option value per cancel event rises monotonically from 30–33 ticks at the touch to 57–68 ticks at the tenth level, reflecting greater adverse selection exposure at depth. Expressed as an expected contribution per order placed, cancel flexibility generates 0.5–3.3 additional ticks per order — a factor of 8–26 times the passive fill return — rising across depths. At levels two and three, the expected value of a fully passive strategy is negative: cancel flexibility is existentially necessary for profitability at these depths.

Repositioning adds substantial alpha at the touch (+585–654 bps) but destroys value at level ten (−85 to −90 bps), identifying a concrete depth threshold for reposition suppression. The ask and bid sides exhibit a widening asymmetry at depth consistent with asymmetric adverse flow patterns. Three algorithmic recommendations follow: level-specific QPR thresholds, reposition suppression at depth, and side-specific signal calibration at levels 7–10.

Future work should extend the counterfactual framework to a cross-section of equities across

liquidity tiers and volatility regimes, apply it over longer samples spanning multiple market conditions, and explore whether learned cancel policies trained on V_{cancel} outcomes outperform the rule-based QPR signal demonstrated here.

Code Availability

All code used to generate the results in this paper is available at: https://github.com/sidladdha/rl_cancel_option_value

References

Foucault, T. (1999). Order flow composition and trading costs in a dynamic limit order market. *Journal of Financial Markets*, 2(2), 99–134.

Glosten, L. R., and Milgrom, P. R. (1985). Bid, ask and transaction prices in a specialist market with heterogeneously informed traders. *Journal of Financial Economics*, 14(1), 71–100.

Goettler, R. L., Parlour, C. A., and Rajan, U. (2005). Equilibrium in a dynamic limit order market. *Journal of Finance*, 60(5), 2149–2192.

Hasbrouck, J. (1991). Measuring the information content of stock trades. *Journal of Finance*, 46(1), 179–207.

Joshi, D., Chopra, S., and Gupta, M. (2025). Reinforcement-learning in limit orders. Working paper.

Kwan, A., and Philip, R. (2025). Reinforcement learning in a dynamic limit order market. NYU Stern Microstructure Meeting 2025 Working Paper.

O'Hara, M. (2015). High frequency market microstructure. *Journal of Financial Economics*, 116(2), 257–270.

Parlour, C. A. (1998). Price dynamics in limit order markets. *Review of Financial Studies*, 11(4), 789–816.

Roşu, I. (2009). A dynamic model of the limit order book. *Review of Financial Studies*, 22(11), 4601–4641.

Pipeline Parameters

Table 4 reports all key configurable parameters governing the cancel signal, order lifecycle, and mid-price buffer.

Table 4: Key Pipeline Parameters

Parameter	Default	Description
`BUFFER_NS`	2 s (2×10^9 ns)	Mid-price ring buffer lookback. Increased from 200 ms to eliminate NaN lookups during sparse sessions.
`MAX_CONCURRENT`	100	Max simultaneous standard+shadow order pairs.
`MAX_LIFETIME_NS`	5 min (3×10^{11} ns)	Max order lifetime before forced cancellation. Fires in ∼4% of L10 events.
`QPR_THRESHOLD`	2.0	QPR cancel gate 1. Recommend 5.0 for L2, 3.0 for L3.
`QPR_QFRAC_GATE`	0.20	Queue-fraction gate 2. Only orders with <20% of queue ahead are eligible for QPR cancel.
`QPR_PERSIST_NS`	10 ms	Persistence gate 3. QPR must be continuously above threshold for 10 ms to fire.
`QPR_WINDOW_NS`	50 ms	Rolling QPR computation window.
`REPOSITION_TICKS`	10 ticks	BBO distance triggering reposition. Recommend ≥15 for levels ≥7.
`warmup_minutes`	10 min	Session warm-up before first virtual order placed.

All timing parameters in nanoseconds unless stated.

Fill Outcome Case Definitions

For each filled standard order, the pipeline assigns a case label (A–D) based on whether the standard and shadow orders ultimately filled and at what price. Table 5 summarises the four cases and their frequencies at levels 1 and 10. The dramatic shift in case composition — from 87% case D (passive) at L1 to 54% case A (repositioned) at L10 — drives the sign reversal in reposition alpha shown in Table 3.

Table 5: Fill Outcome Cases: Frequencies at L1 and L10

Case	Std fills?	Shad. fills?	Freq. L1	Freq. L10	Interpretation
A	Yes	No	12–13%	**50–54%**	Reposition succeeded — original price too far for passive fill.
B	Yes	Yes	∼0%	∼4%	Priority loss — shadow filled at better price; standard at worse repositioned price.
C	No	No	<1%	<1%	Neither fills; reposition contribution zero.
D	No	Yes	86–87%	∼42%	Baseline passive fill — no reposition. Reported as passive return in Table 3.

Case B grows from ∼0% at L1 to ∼4% at L10, consistent with max-lifetime cancels (5-minute cap) at extreme depth. Case A exceeding 50% at L10 signals structural churn rather than selective repositioning.

Reconciliation with Kwan and Philip (2025)

The cancel contribution metric in Kwan and Philip [2025] is defined as:

$$\text{Cancel Contribution } \% \;=\; \frac{\mathbb{E}[\Delta V(s)]}{\mathbb{E}[V_{\text{uncon}}(s)]} \times 100 \tag{C.1}$$

where $\Delta V(s) = V_{\text{uncon}}(s) - V_{\text{con}}(s)$ is the increment to the discounted value function from having cancel optionality available at state s, and $V_{\text{uncon}}(s)$ is estimated via value iteration with $\gamma = 0.99$ at an absorbing-event horizon. At a one-step snapshot horizon, $\mathbb{E}[V_{\text{uncon}}]$ is small (most rewards are approximately zero over a single snapshot interval), inflating the cancel contribution ratio. The replication study Joshi et al. [2025] observes ratios of 256% for GOOGL and 5,200% for AAPL for this reason.

At an absorbing-event horizon (reward computed to first fill, price cross, or cancel), $\mathbb{E}[V_{\text{uncon}}]$ is substantially larger, and the ratio converges toward the original paper's ∼19% estimate for GOOGL. Our per-order contribution framing (equation 4) uses $\mathbb{E}[\text{fill_return}]$ as denominator rather than V_{uncon}, which is why our enhancement ratios (8–27 times) are larger in magnitude than the ∼19% absorbing-event estimate. The two metrics measure different but complementary quantities: Kwan and Philip [2025] measures the fraction of multi-period state value

attributable to cancel optionality; our ratio measures the per-order-placed enhancement of immediate returns from cancel flexibility. Both are internally consistent.